

Question 1

Sigmoid Function

The Logistic function is defined as:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

Compute the values of the sigmoid function for $z = -2, -1, 0, 1, 2$. Show all steps of your calculation.

Solution: The Logistic function is defined as:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

For $z = -2, -1, 0, 1, 2$, the calculated values are:

$$\sigma(-2) = 0.119$$

$$\sigma(-1) = 0.268$$

$$\sigma(0) = 0.500$$

$$\sigma(1) = 0.732$$

$$\sigma(2) = 0.881$$

Question 2

Log-Odds and Probability

1. If the odds of an event are 4 : 1, calculate the corresponding probability and log-odds.
2. If the log-odds increase by 2 units, determine the new odds and probability of the event.

Solution: Given the odds 4 : 1:

- **Probability:**

$$P = \frac{\text{Odds}}{1 + \text{Odds}} = \frac{4}{5} = 0.8$$

- **Log-Odds:**

$$\text{Log-Odds} = \ln(4) \approx 1.386$$

- **After Increasing Log-Odds by 2:**

$$\text{New Log-Odds} = 1.386 + 2 = 3.386$$

$$\text{New Odds} = e^{3.386} \approx 29.55$$

$$\text{New Probability} = \frac{29.55}{1 + 29.55} \approx 0.967$$

Question 3

Likelihood Calculation

Consider a logistic regression model:

$$P(y = 1|x) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \beta_2 x_2)}}$$

Given the model parameters $\beta_0 = -1.5$, $\beta_1 = 0.8$, and $\beta_2 = 0.5$, calculate the predicted probability $P(y = 1)$ for the following data points:

$$(x_1, x_2) = (2.5, 1.3), (x_1, x_2) = (0.5, 2.0), (x_1, x_2) = (1.5, 1.8)$$

Show all intermediate steps.

Solution: The logistic regression model is given by:

$$P(y = 1|x) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \beta_2 x_2)}}$$

Using $\beta_0 = -1.5$, $\beta_1 = 0.8$, and $\beta_2 = 0.5$:

- For $x_1 = 2.5, x_2 = 1.3$, $z = 1.15$, $P(y = 1) \approx 0.759$
- For $x_1 = 0.5, x_2 = 2.0$, $z = -0.1$, $P(y = 1) \approx 0.475$
- For $x_1 = 1.5, x_2 = 1.8$, $z = 0.6$, $P(y = 1) \approx 0.645$

Question 4

Calculating Log-Likelihood

You are given a dataset with the following data points:

Feature (x)	Label (y)
1.0	0
2.0	1
3.0	1

Assume the logistic regression model is defined as:

$$P(y = 1 | x; \beta) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x)}}$$

where $\beta_0 = -1$ and $\beta_1 = 0.5$.

Tasks:

- Calculate the probability $P(y = 1)$ for each data point.
- Compute the log-likelihood for the given dataset.

Solution:

The logistic regression model is:

$$P(y = 1 | x) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x)}}$$

Given: $\beta_0 = -1$, $\beta_1 = 0.5$, and data points $x = [1.0, 2.0, 3.0]$ with $y = [0, 1, 1]$.

(a) Probabilities $P(y = 1)$:

$$P_1 = \frac{1}{1 + e^{-(-1+0.5 \cdot 1)}} = \frac{1}{1 + e^{-(-0.5)}} \approx 0.3775$$

$$P_2 = \frac{1}{1 + e^{-(-1+0.5 \cdot 2)}} = \frac{1}{1 + e^0} = 0.5$$

$$P_3 = \frac{1}{1 + e^{-(-1+0.5 \cdot 3)}} = \frac{1}{1 + e^{0.5}} \approx 0.6225$$

(b) Log-likelihood:

The log-likelihood for logistic regression is:

$$L = \sum_{i=1}^n (y_i \log(P_i) + (1 - y_i) \log(1 - P_i))$$

Substituting the values:

$$L = (0 \cdot \log(0.3775) + 1 \cdot \log(1 - 0.3775)) + (1 \cdot \log(0.5) + 0 \cdot \log(1 - 0.5)) + (1 \cdot \log(0.6225) + 0 \cdot \log(1 - 0.6225))$$

$$L = \log(0.6225) + \log(0.5) + \log(0.6225)$$

$$L \approx -0.4741 - 0.6931 - 0.4741 = -1.6413$$

Question 5

Decision Boundary

Assume a logistic regression model with parameters $\beta_0 = 0.5$ and $\beta_1 = -0.75$. The decision boundary is defined as the point where $P(y = 1) = 0.5$.

Tasks:

- a) Derive the equation for the decision boundary.
- b) Find the value of x where the decision boundary occurs.

Solution:

(a) Decision boundary equation:

The decision boundary occurs when $P(y = 1) = 0.5$:

$$\frac{1}{1 + e^{-(\beta_0 + \beta_1 x)}} = 0.5$$
$$\beta_0 + \beta_1 x = 0 \quad \Rightarrow \quad x = -\frac{\beta_0}{\beta_1}$$

(b) Value of x :

Given $\beta_0 = 0.5$, $\beta_1 = -0.75$:

$$x = -\frac{0.5}{-0.75} = 0.6667$$

Question 6

Question 4: Model Prediction

Consider a logistic regression model with parameters $\beta_0 = 0.2$ and $\beta_1 = 1.5$. Given the input data points $x = \{-1, 0, 1, 2\}$:

Tasks:

- Calculate the predicted probabilities for $P(y = 1)$ for each x .
- Classify each data point into $y = 1$ or $y = 0$ using a threshold of 0.7.

Solution:

(a) Predicted probabilities:

Using the logistic regression model:

$$P(y = 1 | x) = \frac{1}{1 + e^{-(0.2 + 1.5x)}}$$

For $x = -1$:

$$P = \frac{1}{1 + e^{-(-1.3)}} \approx 0.2142$$

For $x = 0$:

$$P = \frac{1}{1 + e^{-(0.2)}} \approx 0.5498$$

For $x = 1$:

$$P = \frac{1}{1 + e^{-(1.7)}} \approx 0.8455$$

For $x = 2$:

$$P = \frac{1}{1 + e^{-(3.2)}} \approx 0.9608$$

(b) Classification:

Using a threshold of 0.7:

$$x = -1, P = 0.2142 \Rightarrow y = 0 \quad x = 0, P = 0.5498 \Rightarrow y = 0 \quad x = 1, P = 0.8455 \Rightarrow y = 1 \quad x = 2, P = 0.9608 \Rightarrow y = 1$$

Question 7

Likelihood Function for Logistic Regression

Consider the logistic regression model for binary classification:

$$P(y = 1 | x) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x)}}$$

Given a dataset with the following feature values and labels:

Feature (x)	Label (y)
1.0	0
2.0	1
3.0	0
4.0	1

Task: Write the likelihood function for the parameters β_0 and β_1 based on the dataset. Compute the likelihood for $\beta_0 = 0$ and $\beta_1 = 1$.

Solution:

The likelihood function is the product of the probabilities of observing the data, given the model parameters. For logistic regression, the likelihood is given by:

$$L(\beta_0, \beta_1) = \prod_{i=1}^m P(y_i | x_i)$$

Where $P(y_i | x_i)$ is:

$$P(y_i = 1 | x_i) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_i)}}$$

and

$$P(y_i = 0 | x_i) = 1 - P(y_i = 1 | x_i)$$

For the given dataset, the likelihood function is:

$$L(\beta_0, \beta_1) = \left(1 - \frac{1}{1 + e^{-(\beta_0 + \beta_1 \cdot 1)}}\right) \left(\frac{1}{1 + e^{-(\beta_0 + \beta_1 \cdot 2)}}\right) \left(1 - \frac{1}{1 + e^{-(\beta_0 + \beta_1 \cdot 3)}}\right) \left(\frac{1}{1 + e^{-(\beta_0 + \beta_1 \cdot 4)}}\right)$$

Substituting $\beta_0 = 0$ and $\beta_1 = 1$:

$$L(0, 1) = \left(1 - \frac{1}{1 + e^{-1}}\right) \left(\frac{1}{1 + e^{-2}}\right) \left(1 - \frac{1}{1 + e^{-3}}\right) \left(\frac{1}{1 + e^{-4}}\right)$$

$$L(0, 1) \approx (1 - 0.7311) \times 0.1192 \times (1 - 0.9526) \times 0.0180$$

$$L(0, 1) \approx 0.2689 \times 0.1192 \times 0.0474 \times 0.0180 \approx 0.0000057$$

Thus, the likelihood for $\beta_0 = 0$ and $\beta_1 = 1$ is approximately 0.0000057.

Question 8

Decision Boundary in Logistic Regression

Given the logistic regression model for binary classification:

$$P(y = 1 | x) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x)}}$$

and the following parameters:

$$\beta_0 = -1, \quad \beta_1 = 2$$

Task: Find the decision boundary for this model. That is, find the value of x where $P(y = 1 | x) = 0.5$.

Solution:

The decision boundary occurs when the probability $P(y = 1 | x)$ is equal to 0.5. From the logistic regression model, we have:

$$P(y = 1 | x) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x)}}$$

Set $P(y = 1 | x) = 0.5$:

$$0.5 = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x)}}$$

Solving for x :

$$1 + e^{-(\beta_0 + \beta_1 x)} = 2$$

$$e^{-(\beta_0 + \beta_1 x)} = 1$$

$$-(\beta_0 + \beta_1 x) = 0$$

$$\beta_0 + \beta_1 x = 0$$

Substituting $\beta_0 = -1$ and $\beta_1 = 2$:

$$-1 + 2x = 0$$

$$2x = 1$$

$$x = 0.5$$

Thus, the decision boundary is at $x = 0.5$.

Question 10

Spam Email Detection

A dataset contains email features such as the number of hyperlinks (x_1) and the number of keywords (x_2). The output y is 1 for spam and 0 for not spam. Suppose the logistic regression model is:

$$P(y = 1 \mid x_1, x_2) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \beta_2 x_2)}}$$

Given the parameters:

$$\beta_0 = -2, \quad \beta_1 = 0.5, \quad \beta_2 = 1$$

Determine whether an email with $x_1 = 3$ hyperlinks and $x_2 = 4$ keywords is classified as spam.

Solution: The logistic regression model is:

$$P(y = 1 \mid x_1, x_2) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \beta_2 x_2)}}$$

Substitute $\beta_0 = -2$, $\beta_1 = 0.5$, $\beta_2 = 1$, and the features $x_1 = 3$, $x_2 = 4$:

$$P(y = 1 \mid x_1, x_2) = \frac{1}{1 + e^{-(-2 + 0.5 \cdot 3 + 1 \cdot 4)}}$$

Simplify:

$$P(y = 1 \mid x_1, x_2) = \frac{1}{1 + e^{-(-2 + 1.5 + 4)}} = \frac{1}{1 + e^{-3.5}}$$

$$P(y = 1 \mid x_1, x_2) \approx \frac{1}{1 + 0.0302} \approx 0.97$$

Since $P(y = 1 \mid x_1, x_2) > 0.5$, the email is classified as **spam**.

Question 11

Credit Risk Assessment

A bank uses logistic regression to predict the probability of a customer defaulting on a loan ($y = 1$) based on their income (x_1) and loan amount (x_2). The model is:

$$P(y = 1 \mid x_1, x_2) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \beta_2 x_2)}}$$

Given:

$$\beta_0 = -1.5, \quad \beta_1 = -0.01, \quad \beta_2 = 0.05$$

Predict the probability of default for a customer with an income of \$50,000 ($x_1 = 50$) and a loan amount of \$200,000 ($x_2 = 200$).

Solution: The logistic regression model is:

$$P(y = 1 \mid x_1, x_2) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \beta_2 x_2)}}$$

Substitute the given parameters and features:

$$P(y = 1 \mid x_1 = 50, x_2 = 200) = \frac{1}{1 + e^{-(-1.5 - 0.01 \cdot 50 + 0.05 \cdot 200)}}$$

Simplify:

$$P(y = 1 \mid x_1, x_2) = \frac{1}{1 + e^{-(-1.5 - 0.5 + 10)}} = \frac{1}{1 + e^8}$$

$$P(y = 1 \mid x_1, x_2) \approx \frac{1}{1 + 2980.96} \approx 0.00034$$

Since the probability is very low ($P \approx 0.034\%$), the customer is unlikely to default on the loan.

Question 12

Disease Diagnosis

A logistic regression model predicts whether a patient has a certain disease ($y = 1$) based on their age (x_1) and cholesterol level (x_2):

$$P(y = 1 \mid x_1, x_2) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \beta_2 x_2)}}$$

Parameters are:

$$\beta_0 = -4, \quad \beta_1 = 0.02, \quad \beta_2 = 0.01$$

Calculate the probability of disease for a patient aged 60 ($x_1 = 60$) with a cholesterol level of 240 ($x_2 = 240$).

Solution: The logistic regression model is:

$$P(y = 1 \mid x_1, x_2) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \beta_2 x_2)}}$$

Substitute $\beta_0 = -4$, $\beta_1 = 0.02$, $\beta_2 = 0.01$, $x_1 = 60$, and $x_2 = 240$:

$$P(y = 1 \mid x_1, x_2) = \frac{1}{1 + e^{-(-4 + 0.02 \cdot 60 + 0.01 \cdot 240)}}$$

Simplify:

$$P(y = 1 \mid x_1, x_2) = \frac{1}{1 + e^{-(-4 + 1.2 + 2.4)}} = \frac{1}{1 + e^{-0.4}} = \frac{1}{1 + e^{0.4}}$$

Approximate:

$$P(y = 1 \mid x_1, x_2) \approx \frac{1}{1 + 1.4918} \approx 0.4013$$

Thus, the probability of disease is approximately 40.13%.

Question 13

Medical Diagnosis - Diabetes Prediction

A logistic regression model predicts whether a patient has diabetes ($y = 1$) based on their Body Mass Index (BMI, x_1) and glucose level (x_2). The model is:

$$P(y = 1 \mid x_1, x_2) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \beta_2 x_2)}}$$

Parameters:

$$\beta_0 = -5, \quad \beta_1 = 0.08, \quad \beta_2 = 0.04$$

Calculate the probability of diabetes for a patient with $x_1 = 30$ (BMI) and $x_2 = 150$ (glucose level).

Solution: The logistic regression model is:

$$P(y = 1 \mid x_1, x_2) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \beta_2 x_2)}}$$

Substitute the given values:

$$P(y = 1 \mid x_1 = 30, x_2 = 150) = \frac{1}{1 + e^{-(-5 + 0.08 \cdot 30 + 0.04 \cdot 150)}}$$

Simplify:

$$P(y = 1 \mid x_1, x_2) = \frac{1}{1 + e^{-(-5 + 2.4 + 6)}} = \frac{1}{1 + e^{3.4}}$$
$$P(y = 1 \mid x_1, x_2) \approx \frac{1}{1 + 29.96} \approx 0.032$$

Thus, the probability of diabetes is approximately 3.2%.

Question 14

Fraud Detection in Transactions

A bank uses logistic regression to predict fraudulent transactions ($y = 1$). The model considers the transaction amount (x_1 , in \$1000s) and the number of unusual patterns detected (x_2). The model is:

$$P(y = 1 \mid x_1, x_2) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \beta_2 x_2)}}$$

Parameters:

$$\beta_0 = -3, \quad \beta_1 = 0.6, \quad \beta_2 = 2$$

Determine if a transaction of \$3000 ($x_1 = 3$) with $x_2 = 2$ unusual patterns is likely fraudulent.

Solution: The logistic regression model is:

$$P(y = 1 \mid x_1, x_2) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \beta_2 x_2)}}$$

Substitute the values:

$$P(y = 1 \mid x_1 = 3, x_2 = 2) = \frac{1}{1 + e^{-(-3 + 0.6 \cdot 3 + 2 \cdot 2)}}$$

Simplify:

$$P(y = 1 \mid x_1, x_2) = \frac{1}{1 + e^{-(-3 + 1.8 + 4)}} = \frac{1}{1 + e^{2.8}}$$
$$P(y = 1 \mid x_1, x_2) \approx \frac{1}{1 + 16.44} \approx 0.057$$

Since $P(y = 1 \mid x_1, x_2) < 0.5$, the transaction is unlikely to be fraudulent.

Question 15

Customer Churn Prediction

A company predicts whether a customer will leave the service ($y = 1$) based on their monthly usage in hours (x_1) and the number of complaints filed (x_2). The model is:

$$P(y = 1 | x_1, x_2) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \beta_2 x_2)}}$$

Parameters:

$$\beta_0 = -1, \quad \beta_1 = -0.02, \quad \beta_2 = 0.5$$

Calculate the probability of churn for a customer who uses the service for 50 hours per month ($x_1 = 50$) and has filed 3 complaints ($x_2 = 3$).

Solution: The logistic regression model is:

$$P(y = 1 | x_1, x_2) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \beta_2 x_2)}}$$

Substitute the given values:

$$P(y = 1 | x_1 = 50, x_2 = 3) = \frac{1}{1 + e^{-(-1 - 0.02 \cdot 50 + 0.5 \cdot 3)}}$$

Simplify:

$$P(y = 1 | x_1, x_2) = \frac{1}{1 + e^{-(-1 - 1 + 1.5)}} = \frac{1}{1 + e^{-(-0.5)}}$$

$$P(y = 1 | x_1, x_2) = \frac{1}{1 + e^{0.5}} \approx \frac{1}{1 + 1.6487} \approx 0.3775$$

Thus, the probability of churn is approximately 37.75%.

Question 16

Autonomous Vehicle Obstacle Detection

An autonomous vehicle uses logistic regression to classify objects as obstacles ($y = 1$) or not ($y = 0$). The model uses object size (x_1 , in square meters) and distance from the vehicle (x_2 , in meters):

$$P(y = 1 | x_1, x_2) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \beta_2 x_2)}}$$

Parameters:

$$\beta_0 = -2, \quad \beta_1 = 1.2, \quad \beta_2 = -0.1$$

Predict whether an object with $x_1 = 1.5$ and $x_2 = 20$ is classified as an obstacle.

Solution: The logistic regression model is:

$$P(y = 1 | x_1, x_2) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \beta_2 x_2)}}$$

Substitute the values:

$$P(y = 1 | x_1 = 1.5, x_2 = 20) = \frac{1}{1 + e^{-(-2 + 1.2 \cdot 1.5 - 0.1 \cdot 20)}}$$

Simplify:

$$P(y = 1 | x_1, x_2) = \frac{1}{1 + e^{-(-2 + 1.8 - 2)}} = \frac{1}{1 + e^{-(-2.2)}}$$

$$P(y = 1 | x_1, x_2) = \frac{1}{1 + e^{2.2}} \approx \frac{1}{1 + 9.025} \approx 0.099$$

Since $P(y = 1 | x_1, x_2) < 0.5$, the object is not classified as an obstacle.

Question 17

Determining Weights for a Single Feature

Consider a logistic regression model for predicting whether a student passes an exam ($y = 1$) or fails ($y = 0$) based on their hours of study (x). The logistic regression model is:

$$P(y = 1 | x) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x)}}$$

We have the following data points:

$$\text{Data: } (x, y) = \{(2, 0), (4, 0), (6, 1), (8, 1)\}$$

Determine the weights β_0 and β_1 using the maximum likelihood estimation (MLE) method.

Solution: The log-likelihood function is:

$$L(\beta_0, \beta_1) = \sum_{i=1}^n [y_i(\beta_0 + \beta_1 x_i) - \ln(1 + e^{\beta_0 + \beta_1 x_i})]$$

For $n = 4$ data points:

$$L(\beta_0, \beta_1) = \sum_{i=1}^4 [y_i(\beta_0 + \beta_1 x_i) - \ln(1 + e^{\beta_0 + \beta_1 x_i})]$$

Substituting the data:

$$\begin{aligned} L(\beta_0, \beta_1) = & (0)(\beta_0 + \beta_1 \cdot 2) - \ln(1 + e^{\beta_0 + \beta_1 \cdot 2}) + (0)(\beta_0 + \beta_1 \cdot 4) - \ln(1 + e^{\beta_0 + \beta_1 \cdot 4}) \\ & + (1)(\beta_0 + \beta_1 \cdot 6) - \ln(1 + e^{\beta_0 + \beta_1 \cdot 6}) + (1)(\beta_0 + \beta_1 \cdot 8) - \ln(1 + e^{\beta_0 + \beta_1 \cdot 8}) \end{aligned}$$

The weights β_0 and β_1 are determined by maximizing $L(\beta_0, \beta_1)$. Using numerical optimization techniques (e.g., gradient descent or software like Python or R), the solution is:

$$\beta_0 \approx -4.5, \quad \beta_1 \approx 0.75$$